

Education

Bachelor of Science in Mechanical Engineering, Taylor University

Anticipated Graduation: May 2027

Minors in Mathematics, Physics, and Honors

GPA: 3.90/4.00, Dean's List

Professional Experience

Mechanical Engineering Intern, Quantinuum

May 2026-August 2026

- Worked as a mechanical design engineer, supporting photonics testing/R&D and electronics packaging at leading quantum computing company
- Owned design of mechanical solutions by integrating optical, electrical, and pneumatic components to increase test cadence by 10X and enable advanced photonics testing
- Reduced system downtime, increased testing capabilities, and safeguarded crucial electronic components through design of mechanical components for electronics test setups and commercial system deployment
- Designed cryogenic test rig for thermal strap qualification and characterization of thermal conductance as a function of bolt torque
- Owned development of UHV test rig to verify components for future cryogenic and vacuum systems

Undergraduate Researcher, Engineering Department, Taylor University

February 2026-Present

- Developing thermal management solutions for an orbital robotics system
- Owning development of research group's first comprehensive, system-wide thermal model in MATLAB
- Performing design, analysis, and TVAC testing of thermal louver for deployment in low-earth-orbit
- Researching and defining environmental, thermal, vacuum, and reliability requirements for robotic arm, body, and joint design sub-teams

Mechanical Engineering Intern, Danner

May 2025-August 2025

- Owned mechanical design for mechanical integration of GNC/autonomy hardware and drivetrain subsystem components for off-road utility vehicles at electric vehicle startup
- Developed an in-house mobile battery charger to save manufacturing team ~6 hours a week, participated in its fabrication, and troubleshoot assembly issues
- Researched and selected parts and suppliers for future vehicle iterations
- Created Jira workflow to manage engineering change requests, reducing lead time for engineering department by ~8 hours per week

Student Activities & Leadership

Hovercraft Racing Competition

February 2025-May 2025

- Responsible for design of thrust and lift subsystem components, increasing hovercraft speed by 35-50%, and leading to 2nd place finish in competition
- Collaborated with electrical team members to design mechanical mounts for electrical components and wire-routing solutions
- Led engineering analysis to optimize selection of motor/propellor combination and inform system design
- Troubleshoot electrical and mechanical manufacturing and assembly issues

Engineering & Physics Department Teaching Assistant

August 2024-Present

- Assisting professors with grading and facilitation/set up of labs
- Leading review/teaching sessions as needed
- Courses include University/General Physics I & II, Modern Physics, Statics, and Fluid Mechanics

Honors Guild Leadership Cabinet

September 2023-Present

- Leading coordination of academic events for University faculty and guest speakers
- Served as a peer mentor and teaching assistant for study abroad trip in Italy (J-Term 2025)
- Responsible for meeting with prospective Honors Students and families to share information and resources.

Technical Skills

- *Mechanical Design:* OnShape, Fusion 360, SolidWorks, AutoCAD, GD&T, PLM (Arena)
- *Simulation & Analysis:* Ansys Fluent, Autodesk FEA/CFD, EES, Hand Calcs
- *Programming:* MATLAB/Simulink, C/C++, Python, Mathematica, Excel, Jira
- *Manufacturing & DFM:* 3D Printing, CNC Machining, Welding, Laser/Plasma Cutting, Waterjet, Sheet Metal